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ELECTROSTATIC GENERATION IN DIELECTRIC FLUIDS – THE VISCOELECTRIC EFFECT

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ABSTRACT

Simultaneous energy transfer modes have been known to interact to produce unusual "coupled" effects. This coupling now has its theoretical basis in the concept of entropy production (or dissipation or irreversibility) central to nonequilibrium irreversible thermodynamics. Over the years, many examples of coupled phenomena have been identified and studied (thermoelectricity, electrokinetics, piezoelectricity, and so forth). Electrohydrodynamics (the effect of fluid motion on electric fields and the reverse effect of electric fields on fluid motion) can be explained as a thermodynamically coupled phenomenon characterized by the viscous and electrical properties of a fluid that contain mobile charges at the molecular (e.g., ions) or macroscopic (e.g., dust) levels. This is called the "viscoelectric" effect.

In the first part of this paper we apply the concepts of irreversible thermodynamics to electrohydrodynamic systems to develop the relevant relationships. The second describes experiments carried out with a new type of Couette electrostatic generator. The resulting experimental data is then discussed in light of the coupled phenomenon relations previously developed.

Most researchers to date have analyzed this phenomenon via ion mobility mechanics and double layer electrohydrodynamics. However, in this proposal we propose using a unique energy-based thermodynamic analysis technique to investigate this phenomenon. Since both momentum and energy produce important conservation laws, they often give uniquely different interpretations of the same phenomenon.

In the discussion below, using published correlations for the flow of dielectric fluids, we show that mass and current flows do appear to be thermodynamically coupled, and call this the thermodynamic "viscoelectric" effect (as opposed to the thermodynamics of electrohydrodynamics). In order to demonstrate the generality of this approach we are requesting funds to a) design and construct a new efficient electrostatic generator using nested annular tube geometry to measure all the relevant physical and flow properties, b) compute the resulting coupling coefficients for various dielectric liquids, c) correlate the coupling coefficients via a dimensionless number, and d) measure the reverse effect (a transverse electric field generating a mass flow or pressure) and compare it with that predicted by the thermodynamic coupled phenomenon theory.

INTRODUCTION

Electrostatic generation has been known since ancient times. It is most commonly observed when certain materials are rubbed together to produce a noticeable electrostatic charge. It has also been known since the middle of the 18th century that electrostatic charging can occur in flowing dielectric fluids. Today this phenomenon is most commonly encountered as an electrical hazard with electrostatic discharges producing unwanted shock, electrical interference, and occasional disastrous explosions in the petroleum, plastics, shipping, aircraft, and agricultural industries [1].

NOMENCLATURE

A = Area
D = Diameter
f = Moody Friction Factor
i = Electric Current
J_i = Generalized Flow
L= Length
L_{ii} = Nonequilibrium Thermodynamic Coefficients
$\dot{m}$ = Mass Flow Rate
p = Pressure

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Q = Volume Flow Rate
T = Absolute Temperature
V= Velocity
X_i = Generalized Force
Φ = Local Entropy Production Rate per Unit Volume
ϕ = Electrical Potential
μ = Dynamic Viscosity
ν = Kinematic Viscosity
ρ = Density
τ_w = Wall Shear Stress
ω = Angular Velocity

BACKGROUND

As Bustin and Dukek [2] point out, while hydrocarbons do not normally ionize appreciably, it only takes one singly ionized impurity particle in 2×10^{12} molecules to produce large electrostatic charging in a moving dielectric fluid. Such low impurity trace concentrations are not easily detectable but the can have a devastating results on the low conductivities of these fluids. In the area of hydrocarbons, very large electrostatic charges have been know to be developed in fuel transport vehicles, refueling of aircraft, filling of fuel storage tanks, filtering, and washing of fuel shipping tanker compartments.

The production of excessive electrostatic charging in the petroleum, aircraft, and combustible dust areas can cause serious explosion and shock hazards. There are numerous reports of exploding fuel storage tanks being filled and fuel tankers being cleaned due to an apparent discharge arc forming between the fuel and an unbonded conductor within the container. Also, a helicopter may carry as an electrostatic potential of as much as 100,000 volts, and anyone touching it before it has been grounded during a landing operation will be seriously injured by the electrical discharge through their body.

The manufacturing industry also suffers from a high incidence of electrostatic charging. While electrostatic technology can sometimes be useful (e.g., printing, electrostatic painting, and power coating), it is most often a serious hazard. During the processing or finishing of dielectric materials (paper manufacture and processing, synthetic textiles, glass, wood, plastics and other chemicals), electrostatic charging is quite common due to surface rubbing and separation.

The most frequent form of electrostatic charging is called “contact charging,” and occurs at the molecular level at an interface of dissimilar materials. The development of a large electrostatic potential requires the physical separation of the materials, one of which must be dielectric. Typical examples are: a hydrocarbon fluid flowing out of a metal pipe or into a metal vessel, film or paper moving across a conductive web or roller, synthetic fabric rubbing on a human, adhesive tape being applied or removed from a conductor, plastic pellets filling a metal hopper, and so forth. While much less of an electrostatic hazard, “inductive charging” can also occur, as in the water-drop electrostatic generator first described by William Thomson (Lord Kelvin) [3].

IRREVERSIBLE THERMODYNAMICS

The study of irreversible thermodynamic phenomena has a curious history beginning with the work of Thomson and Clausius [4]. Various measures of process irreversibility were introduced over the years, but good agreement between theory and experiment was obtained when the product of the local absolute temperature (T) and the local entropy production rate per unit volume (Φ) was used to gage the irreversibility. Additionally, when $T\Phi$ was evaluated for a process, it was always empirically found to result in a sum of "generalized flows" J_i multiplied by their corresponding "generalized forces" X_i, or

$$T\Phi = \Sigma J_i X_i \tag{1}$$

By the middle of the 19th century it had become apparent that many of the empirical relations between the generalized flows and generalized forces for simple phenomena could be modeled as linear, i.e., $J_i = LX_i$, where L was a simple constant of proportionality (e.g., Ohm's and Fourier's laws). As secondary (or coupled) effects were discovered and studied, they too were often found to be linearly related to the generalized forces present. Consequently, numerous linear relations for complex coupled phenomena, based solely on empirical evidence, were in vogue by the end of the 19th century. They were expressed by the following general mathematical model:

$$J_i = \Sigma L_{ij} X_j \tag{2}$$

where the L_{ii} are the "primary" coefficients and L_{ij} ($i \neq j$) are the "secondary" or coupling coefficients. Careful experiments devoted to determining the coefficients L_{ij} often produced the curious result that $L_{ij} = L_{ji}$. In fact, this relation had been proposed in 1851 by Stokes, but without any sound justification [4]. By 1854, Thomson had discovered by empirical means that the coupling coefficient matrix was symmetric for the thermoelectric effect, but could not theoretically justify it except for completely reversible processes; the thermoelectric effect, however, was not a reversible process. Also, early experiments on the conduction of heat in anisotropic crystals produced a coupled effect in that the flow of heat down one axis produced temperature gradients along the other axes.

An adequate theory of coupled phenomena did not emerge until nearly eighty years later when Onsager [5], taking his cue from the behavior of simultaneous chemical reactions and using the concept of microscopic reversibility along with the newly developed techniques of statistical mechanics, succeeded in developing a formal argument establishing the symmetry of the coupling coefficient matrix (i.e., $L_{ij} = L_{ji}$). This contribution was considered to be of monumental importance (Miller [6] suggested it as the fourth law of thermodynamics), and it subsequently earned Onsager the Nobel Prize in Chemistry in 1968.

During the 1940's, Meixner [7], Casimir [8], Prigogine [9], Callen [10] and others established a unified structure for the subject that clearly related entropy production (or dissipation or irreversibility) to the operational generalized forces and flows. The validity of Onsager's reciprocity assertion is always subject to experimental scrutiny, and in 1960 Miller [6] reviewed a massive amount of experimental data available in the literature at that time to confirm its accuracy.

THE VISCOELECTRIC EFFECT

It has been known since the early 1600's that an electric field could exert a force on a dielectric fluid, and by the middle of the next century, it was discovered that the flow of dielectric liquids could generate high voltage (electrostatic) effects (Pickard, [11]). These effects have always been presented as

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curious empirical facts without a unifying theoretical basis. In particular, the generation of static electricity by a flowing liquid or powder has been represented as a "triboelectrification" effect. These effects were first seen as an explosion hazard in the petroleum industry as a result of pipeline and storage tank operations in the early 19th century, and became a serious hazard after 1950 [2, 12, 13]. In recent times, explosive electrostatic conditions have been produced by the flow of dielectric petroleum products during the filling of marine oil tankers [2] and the refueling of aircraft [14]. Air is also a good dielectric and, consequently, is the source of numerous motion generated electrostatic hazards. The generation of atmospheric lightning due to mesoscale circulation of air containing water droplets [15, 16], snow [17, 18], sand and volcanic dust [19] is well known. Moving aircraft and helicopter rotors [20, 21] and spacecraft [22] also produce well-known electrostatic hazards resulting from atmospheric air moving over solid objects. Since 1979, there has been increasing interest in dust explosions (in grain products, plastics, metals, etc.) caused by electrostatic generation and discharge [19, 23]. In addition, the washing of cargo tanks on marine chemical tankers (especially petroleum) with water sprays has been attributed as the source of several tanker explosions due to electrostatic generation in the tank by the motion of the water spray [24-27]. Also, in the late 1970's, detrimental static electrification produced by the flow of oil and other dielectric fluids used for cooling and insulation in transformers began to be a source of concern in the power system equipment industry [28]. Flow-induced electrification was found to affect the dielectric integrity of these fluids at critical points in these systems, occasionally causing catastrophic failures [29].

In the past century, numerous researchers have shown that the reverse effect also exists in which a high voltage electric field applied to flowing dielectric liquid alters its effective viscosity (e.g., [30-32]. This effect is similar to that discovered by Winslow in 1945, wherein a dielectric liquid containing a high concentration of small conductive particles demonstrated dramatic changes in viscosity when subjected to a transverse electric field. When a substantial amount (up to 50% by volume) of fine particles of a conductive solid phase is added to a dielectric liquid they become macroscopically oriented with the applied electrostatic field, thus producing resistance to flow and increasing the effective viscosity of the mixture. This is called the "electrorheological" effect and has received much attention in recent years (viz., Stangroom [33]).

The term "electrohydrodynamics" is used today to describe the phenomena associated with the conversion of electrical energy into kinetic energy and vice versa. For example, electrostatic fields can create hydrostatic pressure (or motion) in dielectric fluids and, conversely, a flow of dielectric fluid in an electrostatic field can produce a potential difference (e.g., see Castellanos [34]; Melcher and Taylor [35]). This is essentially the same as the viscoelectric effect described in this proposal.

We will show below that a thermodynamic coupling exists between the mass flow of a dielectric liquid and an electrostatic field normal to the flow direction. We have chosen the name "viscoelectric" to describe this effect which is consistent with the term "thermoelectric" in which the dominant phenomena (thermal or viscous) is put before the secondary (electric) phenomena.

DEVELOPMENT OF THE THEORY

Here we develop a nonequilibrium thermodynamics theory for fluid electrodynamics based on a two-flow model: 1) fluid (mass) flow and 2) electron flow[1]. The corresponding linearly coupled nonequilibrium thermodynamics flux equations are:

$$J_{current} = i/A_i = L_{\phi\phi}\nabla\phi + L_{\phi m}\nabla(p/\rho) \tag{3}$$

and

$$J_{mass} = \dot{m}/A_m = L_{m\phi}\nabla\phi + L_{mm}\nabla(p/\rho) \tag{4}$$

where i is the electrostatic electrical current, $\dot{m}$ is the mass flow rate of the fluid, A_i and A_m are the current and mass flux cross sectional areas, $\nabla\phi$ is the electrical energy gradient normal to the direction of flow, and $\nabla(p/\rho)$ is the pressure energy gradient in the direction of flow. $L_{\phi\phi}$ and L_{mm} are the primary coefficients obtained from Ohm's[2] and Bernoulli's laws, and (assuming reciprocity holds) $L_{\phi m} = L_{m\phi}$ is the secondary, or coupling, coefficient. In the SI system, these quantities have the following units.

Quantity	Units
$J_{current}$	A/m^2
J_{mass}	$kg/(s \cdot m^2)$
$\nabla\phi$	$V/m = kg \cdot m/(s^3 \cdot A)$
$\nabla(p/\rho)$	$N/kg = m/s^2$
$L_{\phi\phi}$	$A^2 \cdot s^3/(m^3 \cdot kg)$
L_{mm}	$kg \cdot s/m^3$
$L_{\phi m}$	$A \cdot s^2/m^3$
$L_{m\phi}$	$A \cdot s^2/m^3$

The first indication that the viscoelectric effect could be described as a nonequilibrium thermodynamic coupled phenomenon came from an inspection of the empirical correlation developed by Koszman and Gavis [36, 37] for the transverse electrostatic current produced in the pipeline flow of a dielectric hydrocarbon liquid. Their results in equation form are:

$$i = K_1 V^{1.87} D^{0.87} \tag{5}$$

where K_1 is an empirical constant. However, the Hazen-Williams equation for the axial pressure energy gradient in turbulent pipe flow can be written as (e.g., see Streeter and Wylie [38]):

$$\nabla(p/\rho) = CQ^{1.85}D^{-4.87} \tag{6}$$

where C dimensionless is a constant that depends on the roughness of the pipe wall and $Q = A_m V$ is the volume flow rate of the fluid and D is the pipe diameter. While the Hazen-Williams equation was developed for water flowing at ordinary temperatures, the viscosities of water and petroleum fuels are nearly the same (within an order of magnitude) at these temperatures and therefore it should model the flow behavior of these materials as well.

Note that the flux areas in Eq.s (3) and (4) are both equal to the pipe cross sectional area ($A_i = A_m = A = \pi D^2/4$) because the current flux moves with the entrained fluid. When the electrical potential gradient along the pipe wall is zero, the electron flux of Eq. (3) reduces to $i = AL_{\phi m}\nabla(p/\rho)$, and

[1] Castellanos ([34], pp72-75), using thermodynamic coupled phenomenon theory, discusses the coupling of a thermal gradient and an electric field. The research proposed here focuses on the coupling of a pressure gradient and an electric field.

[2] Note that $L_{\phi\phi}$ is the inverse of resistivity, which is the conductivity, and as SI units of $A^2 \cdot s^3/(m^3 \cdot kg)$ = Siemens/m.

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inserting the Hazen-Williams formula with $Q = AV = \pi D^2V/4$, produces:

$$\begin{aligned} i_{turb.} &= (\pi D^2/4)(L_{\phi m})_{turb.}(C)(\pi D^2V/4)^{1.85}D^{-4.87} \\ &= (L_{\phi m})_{turb.}(\pi/4)^{2.85}CV^{1.85}D^{0.83} \\ &= K_1V^{1.85}D^{0.83} \end{aligned} \quad (7)$$

which is nearly identical to the Koszman and Gavis result (Eq. 5) with $K_1 = (L_{\phi m})_{turb.}(\pi/4)^{2.85}C$. The nonequilibrium thermodynamic theoretical explanation of this effect as coupled phenomenon appears to be qualitatively correct.

COMPUTATION OF THE COUPLING COEFFICIENT

We can now compute the coupling coefficient for turbulent pipe flow of a dielectric liquid from this result as $L_{\phi m} = L_{m\phi} = K_1[(\pi/4)^{2.85}C]^{-1}$, where K_1 is the Koszman and Gavis empirical constant. Using the data from Koszman and Gavis, we can determine an approximate value of K_1 for stainless steel pipes of $K_1 \approx 3\times10^{-8}\ A\cdot s^{1.87}/m^{2.74}$. Using C = 140 (very smooth pipes), then we can calculate $L_{\phi m} = L_{m\phi} \approx 4\times10^{-10}\ A\cdot s^{1.87}/m^{2.74}$. Note that the unit exponents here do not exactly match those in the table above (i.e., the units of $L_{\phi m}$ and $L_{m\phi}$ should be $A\cdot s^2/m^3$ not $A\cdot s^{1.87}/m^{2.74}$). However, several investigators, including Oomen and Lindgren [39], have found the current to increase with the square of the velocity at high flow rates,

This apparent discrepancy is due to the fact that the Hazen-Williams equation is a relatively simple approximation to the pressure gradient - flow rate behavior in pipes over the complete turbulent region. A better approximation is given by the Darcy-Weisbach equation (see Potter and Wiggert [40]),

$$\nabla(p/\rho) = (f/D)V^2/2 = [8f/(\pi^2D^5)]Q^2 \quad (8)$$

where f is the pipe "friction factor," $\nabla(p/\rho)$ is $(\Delta p/\rho L)$, and L and D are the pipe length and diameter. The Moody diagram describes the non-linear relation between friction factor f, Reynolds number $Re = \rho VD/\mu$, and the relative pipe roughness ε/D. Inspection of this diagram reveals that at high Reynolds number the friction factor becomes a constant, producing a condition where $\nabla(p/\rho)$ is proportional to Q^2. Low Reynolds numbers have f varying linearly with Re, producing a condition where $\nabla(p/\rho)$ is proportional to Q. Intermediate Reynolds numbers produce a condition where $\nabla(p/\rho)$ can be approximated as proportional to Q^n/D^m (the Hazen-Williams equation has n = 1.85 and m = 4.87).

The laminar flow of a dielectric liquid in a circular horizontal pipe has been treated analytically by Walmsley and Woodford [41] and experimentally studied by Touchard [42]. Oommen and Lindgren [39] also studied it experimentally in a horizontal annulus. In both cases the current was found to be a linear function of the mean flow velocity V. For laminar flow in a circular horizontal pipe, the pressure gradient is:

$$(\nabla p/\rho)_{laminar} = 32\nu V/D^2 \quad (9)$$

and again taking $\nabla\phi = 0$ in Eq. (3) with $A = \pi D^2/4$ we obtain for the current $i_{laminar} = 8\pi\nu V(L_{\phi m})_{laminar}$. Since the velocity can be defined in terms of the Reynolds number as $V = \nu Re/D$, then the laminar flow electrostatic current can be written as:

$$i_{laminar} = (L_{\phi m})_{laminar}(8\pi\nu^2/D)Re = K_2Re \quad (10)$$

where the constant $K_2 = (L_{\phi m})_{laminar}(8\pi\nu^2/D)$. This is the same linear relation observed by Oommen, and Lindgren for laminar annular flow, and Touchard for laminar pipe flow. Note that we can now compute $L_{\phi m} = K_2[8\pi\nu^2/D]^{-1}$ when K_2 is known.

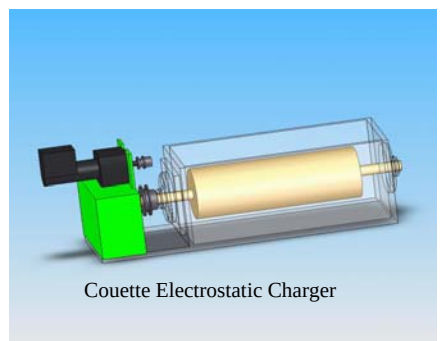
Eq. 3 above appears to hold for both laminar and turbulent flow, while Eq. 4 only appears to hold for laminar flow since $J_{mass} = \dot{m}/A_m = \rho A_mV/A_m = \rho V$, but $\nabla(p/\rho) \propto V^2$ in turbulent flow. Since linear thermodynamic theory is valid only for sufficiently slow processes in systems near equilibrium, this poses a problem unless $(L_{mm})_{turbulent} \propto 1/V$. However, this is in fact the case. A Taylor's series first-order expansion on J_{mass} shows that:

$$L_{mm} = \partial J_{mass}/\partial[\nabla(p/\rho)] \quad (11)$$

and since $J_{mass} \propto V \propto [\nabla(p/\rho)]^{1/2}$ then $\partial J_{mass}/\partial[\nabla(p/\rho)] \propto [\nabla(p/\rho)]^{-1/2} \propto 1/V$. Consequently, the basic concept of modeling viscoelectricity as coupled irreversible thermodynamic phenomena for both laminar and turbulent pipe flow appears to be sound.

EXPERIMENTAL APPROACH

One of the issues in conducting laboratory experimental research in this area is to develop a stable and reproducible test apparatus that can be easily instrumented. A new type of Couette electrostatic charger has been developed for studying viscoelectric effects in the laboratory [43]. This apparatus consists of a dielectric fluid forced through the annular gap between a pair of vertical concentric cylinders while the inner cylinder is rotated about its axis. Salton [44] developed an alternate version of this apparatus that has a horizontal rotating cylinder partially submerged in a fixed volume of dielectric fluid to be tested. In both cases, the electrostatic current increases linearly with the cylinder's rotational speed so long as the boundary layer flow is laminar. It then becomes nonlinear when the flow field becomes turbulent.


Couette Electrostatic Charger

In pipe flow the wall shear stress τ_w produces the resulting pressure gradient. This is given by ([40], pg 250) $\tau_w/\rho = (D/4L)(\nabla p/\rho)$ and consequently, $\nabla p/\rho = (4L/D)(\tau_w/\rho)$. Therefore, $\nabla p/\rho$ and τ_w/ρ differ only by a constant (4L/D). In the flow between rotating cylinders, with the outer cylinder fixed and $r_0 >> r_i$, ([40] pg 265) $\tau_w/\rho = 2\nu\omega$, where $\nu = \mu/\rho$, the dynamic viscosity, and ω is the rotational velocity of the cylinder. Using τ_w/ρ as the generalized force in this case, it is seen that the coupled current flux should vary linearly with the cylinders rotational velocity until the laminar flow condition is exceeded as noted experimentally above.

In an attempt to develop a reproducible fluid sloshing electrostatic test apparatus, a Bellco Glass Co. a temperature controlled orbital low-speed shaker was used as the test bed. The shaker's stainless steel tank and cover makes it an ideal Faraday cage. The tank sat on three off-set bearings, so that the tank moved in a horizontal circle, 1" in diameter. Tank oscillation speed was variable from 0 to 1.33 cps (hz) and creates both lateral and longitudinal standing waves in the tank fluid.

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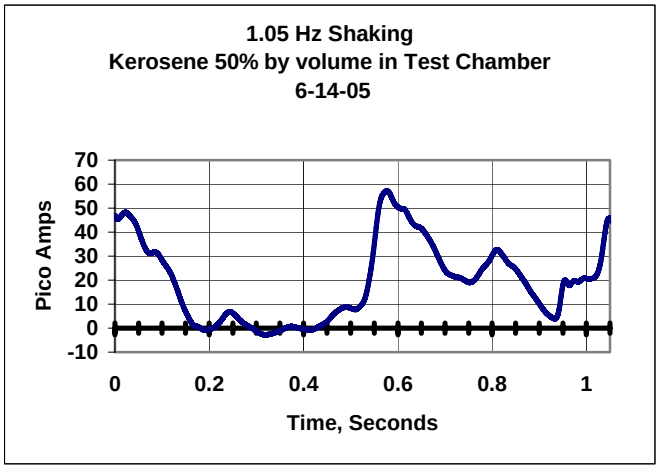
To eliminate triboelectric electrostatic generation in moving wires, the electrometer was mounted on top of the moving covered tank. There was then only a small (1-2 pA) signal due to slight cable movement via their inertia.

The text chambers consisted of carbon steel tubes, 7.6 cm OD with 1.6 mm wall thickness and 30.5 cm long, were sealed on one end. The open end was sealed with a rubber stopper holding an inner electrode. The inner electrode was 8-sided star-shaped, 5.7 cm OD aluminum blades, 3.2 mm thick and 25.4 cm long on an aluminum shaft that penetrated the center of the rubber stopper. The test chambers were placed horizontally in the shaker so that the test fluid sloshed end-to-end.

The tubes were filled 50% by volume (600 ml) with a kerosene-oil mixture having a conductivity of 19 pico siemens per meter. The figure below shows typical results from 1.05 hz shaking.



The wave pattern here is very repeatable, and is the result of the test liquid sloshing in the container. At time 0, the cylindrical test chamber is at the bottom dead center of its movement. The center peak corresponds to the container position 180 degrees later (top dead center). The wave pattern varies greatly with the shaker frequency, as does the internal fluid motion.

CONCLUSIONS

This research establishes that the electrohydrodynamic phenomenon of electrostatic energy generation in moving dielectric fluids using can be understood via nonequilibrium thermodynamic coupled phenomenon theory. Researchers currently analyze this phenomenon via classical mechanics (i.e., conservation of momentum), however we use a thermodynamics (conservation of energy) approach. The application of this technique is new to this subject, but has been successfully used in the past to explain similar phenomena such as electrokinetics, thermoelectricity, and so forth. Data in the literature suggests that a thermodynamically coupled current-mass flux relation exists for both laminar and turbulent pipe flow. Extensive data in the electrohydrodynamics literature demonstrates that the reverse viscoelectric effect also exists; consequently we propose to model both flow-electrostatic field interactions in electrohydrodynamics through the use of nonequilibrium thermodynamic theory.

This phenomenon has a significant group of industrial applications (painting, printing, air purification, etc.), but it is most commonly encountered today as a serous electrical hazard with electrostatic discharges causing unwanted shock, electrical interference, and occasional disastrous explosions in the petroleum, plastics, shipping, aircraft, and agricultural industries,. Tubular flow, Couette, and orbital shakers can produce reproducible electrostatic charging data on relatively small dielectric fluid samples. Successfully modeling this phenomenon will allow future designers to better understand and prevent unwanted viscoelectric shock hazards and to investigate electrostatic propulsion systems.

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